List of Publications

Journals

[1] Utkalika Satapathy et al. "XPLOG: A Dynamic Observability Framework for Distributed Sandboxed Microservices". In: *IEEE Transactions on Services Computing* (-). Pre-print. DOI: `10.1109/TSC.2025.3612900`.

[2] Shubha Brata Nath et al. "Containerized Deployment of Micro-services in Fog Devices: A Reinforcement Learning-based Approach". In: *The Journal of Supercomputing (JSUP), Springer* 78.5 (2022), pp. 6817–6845. DOI: `https://doi.org/10.1007/s11227-021-04135-2`.

[3] Subhrendu Chattopadhyay et al. "Aloe: Fault-Tolerant Network Management and Orchestration Framework for IoT Applications". In: *IEEE Transactions on Network and Service Management* 17.4 (2020), pp. 2396–2409. DOI: `10.1109/TNSM.2020.3008426`.

[4] Sandip Chakraborty, Sukumar Nandi, and Subhrendu Chattopadhyay. "Alleviating hidden and exposed nodes in high-throughput wireless mesh networks". In: *IEEE Transactions on Wireless communications* 15.2 (2016), pp. 928–937. DOI: `10.1109/TWC.2015.2480398`.

Conferences

[1] Bratin Mondal et al. "GranuloTrack: Kernel-Deep Telemetry with Event-Driven Precision for System Observability". In: *Under Review*. Under Review. -.

[2] Abhinav Prakash et al. "Chronark: Security-Aware Multi-Cloud Benchmarking for Distributed Microservices". In: *Under Review*. Under Review. -.

[3] Neha Chowdhary et al. "BeeGuard: Explainability-based Policy Enforcement of `eBPF` Codes for Cloud-native Environments". In: *Seventeenth International Conference on COMmunication Systems NETworkS (COMSNETS)*. Vol. 17. Bangalore, IN, June 2025. DOI: `10.1109/COMSNETS63942.2025.10885589`.

[4] Tanmoy Dutta et al. "AutoPAC: Exploring LLMs for Automating Policy to Code Conversion in Business Organizations". In: *Seventeenth International Conference on COMmunication Systems NETworkS (COMSNETS)*. Vol. 17. Bangalore, IN, June 2025. DOI: `10.1109/COMSNETS63942.2025.10885751`.

[5] Arpit Tripathi et al. "MADE: An MEC Supported Platform Independent and Version Agnostic Framework for Mobile Application Deployment". In: *Sixteenth IEEE International Conference on Advanced Networks and Telecommunications Systems (ANTS)*. Vol. 16. Guwahati, IN, Apr. 2024. DOI: `10.1109/ANTS63515.2024.10898186`.

[6] Utkalika Satapathy et al. "DisProTrack: Distributed Provenance Tracking over Serverless Applications". In: *Forty First IEEE International Conference on Computer Communications (INFOCOM)*. Vol. 41. NewYork, US, Apr. 2023. DOI: `10.1109/INFOCOM53939.2023.10228884`.

[7] Subhrendu Chattopadhyay et al. "Amalgam: Distributed Network Control With Scalable Service Chaining". In: *Nineteenth IFIP Networking Conference (IFIP Networking)*. Vol. 19. Paris, FR, 22-25 June 2020, pp. 519–523.

[8] Subhrendu Chattopadhyay et al. "Aloe: An Elastic Auto-Scaled and Self-stabilized Orchestration Framework for IoT Applications". In: *Thirty Eighth IEEE International Conference on Computer Communications (INFOCOM)*. Vol. 38. Paris, FR, 29 April - 2 May 2019. DOI: `10.1109/INFOCOM.2019.8737656`.

[9] Shubha Brata Nath et al. "PTC: Pick-Test-Choose to Place Containerized Micro-Services in IoT". In: *2019 IEEE Global Communications Conference (GLOBECOM)*. Waikoloa, US, Sept. 2019, pp. 1–6. DOI: `10.1109/GLOBECOM38437.2019.9013163`.

[10] Subhrendu Chattopadhyay et al. "Improving MPTCP Performance by Enabling Sub-Flow Selection over a SDN Supported Network". In: *Fourteenth International Conference on Wireless and Mobile Computing, Networking and Communications (WiMob)*. Vol. 14. Limmasol, CY, 15-17 October 2018. DOI: `10.1109/WiMOB.2018.8589143`.

[11] Subhrendu Chattopadhyay et al. "Flipper: Fault-Tolerant Distributed Network Management and Control". In: *Fifteenth IFIP/IEEE International Symposium on Integrated Network Management (IM)*. Vol. 15. Lisbon, PT, Aug. 2017. DOI: `10.23919/INM.2017.7987307`.

[12] Pranav Kumar Singh et al. "Fast and Secure Handoffs for V2I communication in Smart City Wi-Fi Deployement". In: *Fourteenth International Conference on Distributed Computing and Internet Technology (ICDCIT)*. Vol. 14. Bhubaneswar, IN, 13-16 January 2017. DOI: 10.1007/978-3-319-72344-0_15.

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Under Review

[1] Bratin Mondal et al. "GranuloTrack: Kernel-Deep Telemetry with Event-Driven Precision for System Observability". In: *Under Review*. Under Review. -.

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